

Packet-to-cloud Translation Sprint

Part A - User

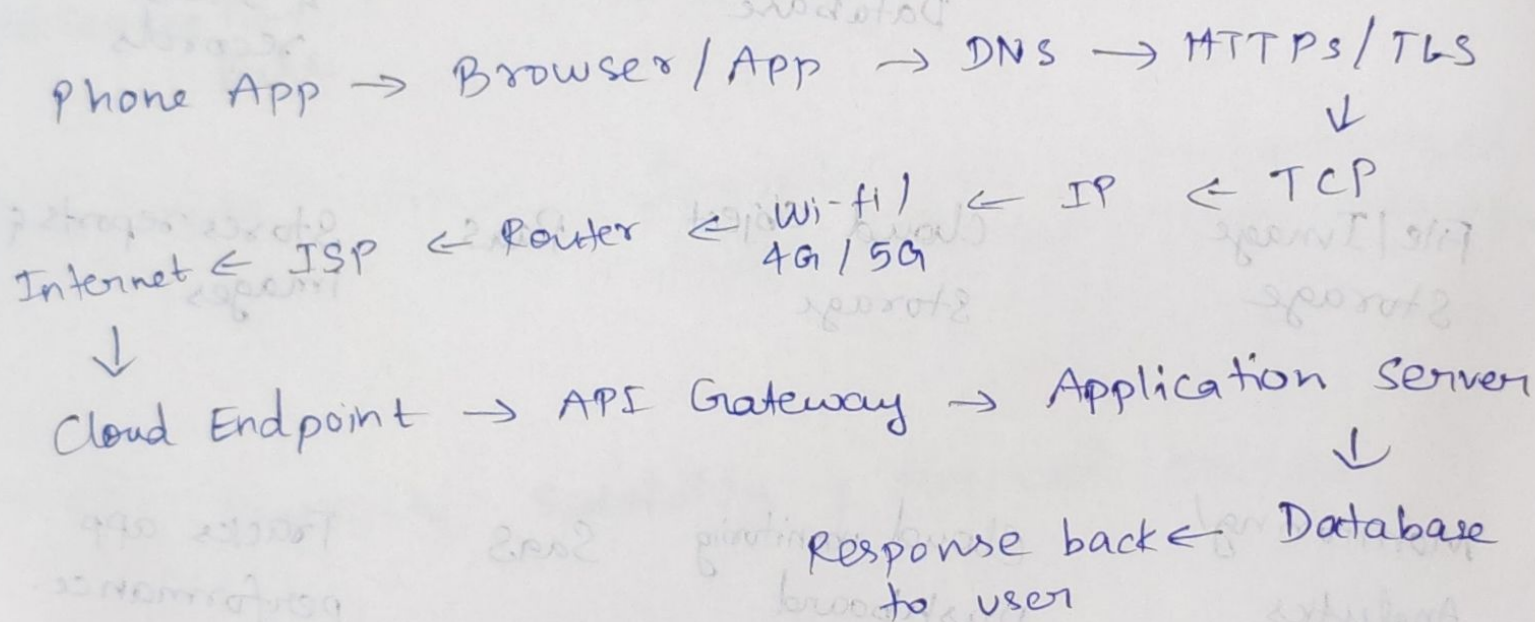
A1 Sleep scope: Student Sleep & Academic Stress Analytics App

⇒ submit sleep hours & stress levels

A2 ⇒ send my sleep hours & stress levels to the cloud & show my stress analysis

Part B -

B1 Message path



B2 Technical Translation Table

Possible failure	Cloud-side protection	Layer / Component	What it does in your app
DNS not found	DNS redundancy	DNS	converts app domain to server IP
Certificate expired	Valid SSL certificate	HTTPS/TLS	encrypts sleep & stress data
Packet loss	Retransmission Transfers data	TCP/IP Router/ISP/ Internet	Transfers data reliably

Part C cloud Landing Zone

cloud Need	Component choice	Service Model	Justification
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User identity / login	Authentication Service	SaaS	Secure student login
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App backend	Cloud Application server	PaaS	Runs analytics logic
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Database	Managed SQL Database	PaaS	Stores sleep & stores records
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File/Image storage	cloud object storage	PaaS	Stores reports & images
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Monitoring/ Analytics	cloud monitoring Dashboard	SaaS	Tracks app performance.
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Part D - API Contract

Field	Answer
Endpoint URL	/api/sleepscope/submit
HTTP Method	POST
Request Data	Student ID, sleep hours, stress level, date
Response Data	Success message, stress score, sleep recommendation
Authentication Rule	JWT token after login
Error Case	Invalid data or unauthorized access returns error message

Part E - Elasticity, Scalability & Measured Service

E1.
$$7200 \text{ requests} \div 10 \text{ minutes} = 720 \text{ requests/minute}$$

E2.
$$720 \times 1.25 = 900 \text{ requests/minute}$$

E3.
$$900 \div 180 = 5 \text{ instances}$$

E4.
Enable auto-scaling so additional cloud servers start automatically during peak usage

E5.

- Request count
- Average response time (latency)

Part F - AI / Google Research Evidence

Search / Prompt used	Tool used	Useful Finding	How I Applied it
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Best cloud architecture for student health analytics app

ChatGPT
Managed databases
& HTTPS
improve security
& reliability

Used HTTPS
API Gateway
and managed database in the app design

These answers are tailored specifically to your SleepScope project & match the requirements of the PDF.